UNIVERSITY OF SCIENCE, VNU-HCM
FACULTY OF INFORMATION TECHNOLOGY

COURSE REPORT

Course: Pattern Recognition
Class: 23TGMT

Instructors: Võ Thanh Lâm
Lê Hoàng Thái
Nguyễn Thanh Tình

Students: Trương Quốc Cường – 23127333
Lê Anh Duy – 23127011
Trần Gia Huy – 23127199

Ho Chi Minh City, 2026

Preface

This report template is prepared for academic writing in English using LaTeX. It includes a title page, table of contents, lists of figures and tables, a glossary, and a bibliography workflow with XeLaTeX and Biber.

During drafting, language-assistance tools such as ChatGPT may be used to improve phrasing and readability [1].

General Information

- **Report Title:** (fill in report title)
- **Course / Unit:** (fill in course or department)
- **Team Leader:** Trương Quốc Cường (23127333)
- **Team Members:** Lê Anh Duy (23127011), Trần Gia Huy (23127199)
- **Objective:** (fill in objective)
- **Scope:** (fill in scope)
- **Tools / Environment:** (fill in tools and environment)

After finalizing the information above, continue with the main content following the chapter/section structure.

Table of Contents

Preface **1**

General Information **2**

List of Figures **4**

List of Used Images **5**

List of Tables **6**

Glossary and Abbreviations **7**

I WEEK 1 **8**

Question 1. Distinguish basic, applied, and development-oriented research 8

Question 2. Example application in pattern recognition 8

Question 3. Supervised, unsupervised, and reinforcement learning 8

Question 4. Why meta-learning is useful 9

II WEEK 2 **10**

References **12**

List of Figures

II.1 Sample generated image for demonstration 11

List of Used Images

This section lists images inserted directly with `\includegraphics` and where they are used in the document.

- `assets/HCMUS.png`: university logo on the cover page (in `cover.tex`).
- `assets/F2.jpg`: sample generated image in Week 2 (Figure II.1).

List of Tables

1 Abbreviations 7
2 Terms 7

I.1 Supervised learning vs meta-learning by task type 9

Glossary and Abbreviations

A. Abbreviations

Table 1: Abbreviations

Abbrev.	**Meaning**	**English Expansion**
API	Application programming interface	Application Programming Interface
CNN	Convolution neural network	Convolutional Neural Network
GPU	Graphics processor	Graphics Processing Unit
ML	Machine learning	Machine Learning
NLP	Language processing by computers	Natural Language Processing
OCR	Optical text recognition	Optical Character Recognition
PSNR	Peak signal-to-noise ratio	Peak Signal-to-Noise Ratio
RL	Learning by reward	Reinforcement Learning
SSIM	Structural similarity metric	Structural Similarity Index
VLM	Vision-language model	Vision-Language Model

B. Terms

Table 2: Terms

Term Used in Report	**Reference Term**
Baseline model	Baseline
Domain shift	Domain shift
Feature extraction	Feature extraction
Few-shot learning	Few-shot learning
Fine-tuning	Fine-tuning
Generative model	Generative model
Inference latency	Latency
Meta-learning	Meta-learning
Prompt engineering	Prompt engineering
Support set	Support set

Chapter I

WEEK 1

Question 1. Distinguish basic, applied, and development-oriented research

In computer vision and pattern recognition, these three tracks usually form a pipeline: foundations, practical solutions, and product deployment.

Basic research

- Focus: new theory, algorithms, and analysis.
- Output: validated method and empirical study.
- Typical metrics: accuracy, robustness, complexity.

Applied research

- Focus: solve a concrete real-world problem in a domain.
- Output: task-oriented pipeline and measurable KPI improvement.
- Typical constraints: noisy data, domain shift, limited labels.

Development-oriented research

- Focus: run models reliably in production.
- Output: scalable service, monitoring, and release workflow.
- Typical constraints: latency, cost, reliability, maintainability.

Question 2. Example application in pattern recognition

A retail shelf-monitoring system detects products, tracks stock level, and alerts out-of-stock events. Evaluation combines detection quality with business metrics such as false alerts and response time.

Question 3. Supervised, unsupervised, and reinforcement learning

Supervised learning

Uses labeled data and optimizes prediction loss directly.

Unsupervised learning

Finds structure in unlabeled data, for example clustering and representation learning.

Reinforcement learning

Learns a policy from reward signals through interaction with an environment.

Question 4. Why meta-learning is useful

Meta-learning improves adaptation speed when new classes or tasks have very few examples. It is especially useful when annotation is expensive and task distribution changes often.

Compact comparison

Table I.1: Supervised learning vs meta-learning by task type

Task	**Supervised Learning**	**Meta-Learning**
Image classification	Needs many labeled samples per class	Adapts to new classes with few-shot support
Object detection	High annotation cost for bounding boxes	Better adaptation for new categories/domains
Image segmentation	Pixel labels are expensive to create	Few-shot segmentation with small support set

Chapter II

WEEK 2

Question 1: What is a generative model?

A generative model learns a data distribution and can synthesize new samples that follow similar structure to training data.

Question 2: Business value of Generative AI

Common revenue channels include:

- Subscription plans for end users and teams.
- API-based usage billing integrated into enterprise products.

Question 3: Practical image/video tool example

Generated media can support marketing, prototyping, content localization, and design iteration with lower turnaround time.

Question 4: Compare VAE, Flow, GAN, and Diffusion

Family	**Core mechanism**	**Image quality**	**Sampling speed**	**Training stability**
VAE	Encode/decode through latent variables	Medium to high	Fast	High
Flow	Invertible transformation from noise	High	Fast	Medium to high
GAN	Adversarial generator/discriminator game	High	Fast	Medium
Diffusion	Iterative denoising from Gaussian noise	Very high	Slower	High

Question 5: How diffusion models generate images

A diffusion model adds noise during training and learns the reverse denoising process. At inference, it starts from random noise and repeatedly denoises until a clean image appears.

Question 6: Compare popular diffusion-based systems

Closed systems typically optimize quality and user experience, while open systems (such as open-weight families) prioritize customization.

Question 7: Example output image

Figure II.1: Sample generated image for demonstration

Question 8: Practical multimodal use case

A document assistant can combine OCR, layout understanding, and language modeling to answer questions from scanned invoices and forms.

References

[1] OpenAI. "Chatgpt," Accessed: Jan. 25, 2026. [Online]. Available: https://chat.openai.com/